Hazard Mitigation Plan

Alameda County, California

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Alameda County, California

Alameda County is located in the East side of San Francisco's Bay Area. This is a highly populated region, home to over 1.6 million people (US Census Bureau Table, 2023). Alameda County contains 14 incorporated cities and six unincorporated communities and rural areas throughout its 813 square miles, as seen in Figure 1.1. Of the Alameda County residents, 34.5% identify as Asian, 27.9% White, 23.3% Hispanic, 10.5% Black, and 3.8% other (US Census Bureau Table, 2023). The majority of this diverse community lives close to the East shore of the San Francisco Bay and along the Hayward fault. Proximity to this body of water and ever moving fault line has the potential to endanger the communities of Alameda County in the event of a natural hazard like an earthquake. This Hazard Mitigation Plan identifies risks, analyzes proposed goals and policy objectives, and recommends actions in order to preserve the quality of life and support the economy of the Alameda County community. According to FEMA, hazard mitigation planning reduces loss of life and property damage by abating the disaster (Seismic Building Codes, 2023). Mitigation planning is a preemptive measure to limit impacts from local hazards. This plan identifies communities at higher risk, as well as the overall risk to the residents and infrastructure of the county, to develop and implement adjusted mitigation strategies. It is important to gain a full understanding of a community's exposure, sensitivity, and adaptive capacity to a natural hazard to recommend changes. The purpose of this chapter is to recognize earthquakes as a critical natural hazard and identify vulnerable communities within Alameda County, California.

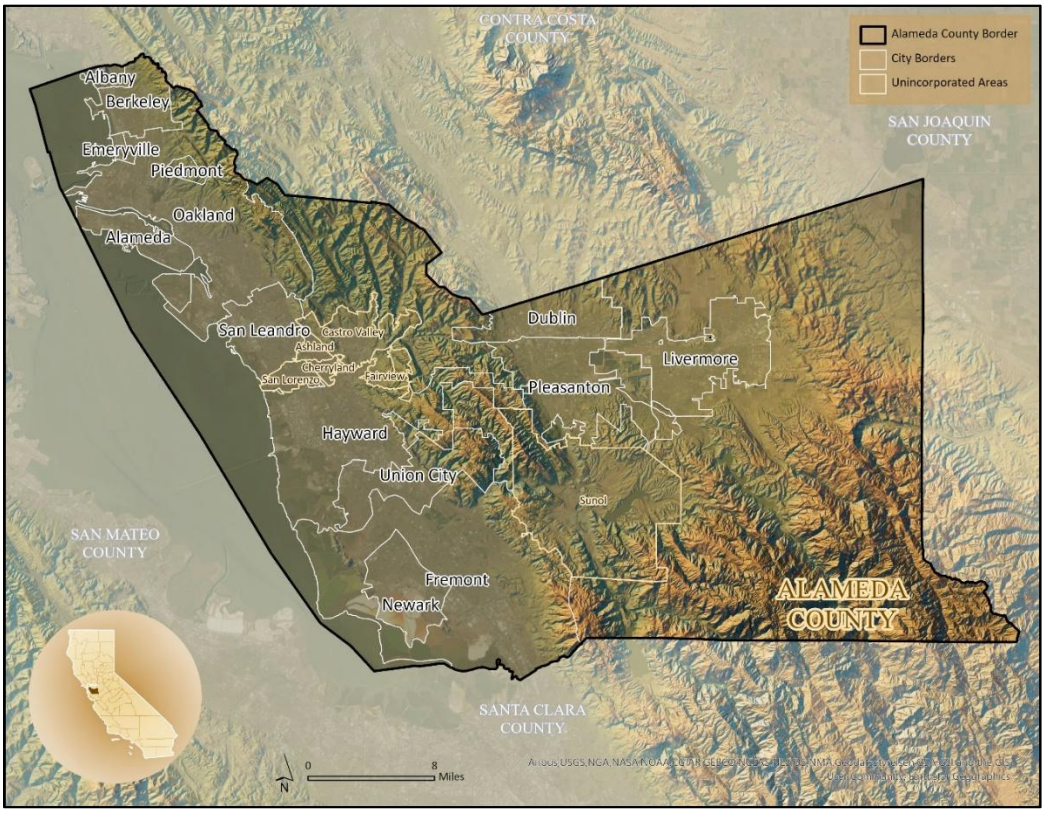


***Figure 1.1**: Alameda County*

Hazard Exposure

A community's exposure level to a natural hazard increases or decreases their vulnerability and therefore risk of the hazard. Proximity to a seismic event is the most significant aspect of exposure. Alameda County's proximity to the Hayward fault categorizes a 4.0-5.0 magnitude earthquake or above as a hazard because buildings and human lives are at greater risk (USGS Frequently Asked Questions, 2023). When seismic activity begins, the buildings and people experience shaking first, but if the intensity and duration increase, severity of damage dramatically rises. Most faults running along California are strike slip faults, which have historically generated earthquakes less than magnitude 8.0, in comparison to the potential magnitude > 9.0 at the Cascadia subduction zone. Other major faults exist in the Bay Area, but the Hayward fault runs directly through the cities of Berkeley, Oakland, San Leandro, Hayward, Union City, and Fremont. The last major earthquake caused by the Hayward fault was in 1868 with a 6.8 magnitude shake (Brocher et al., 2018). The previous five earthquakes occurred at approximately 150-year intervals. According to this pattern, in 2025 the Hayward fault is seven years overdue (Brocher et al., 2018). The Calaveras fault (San Ramon segment), which crosses through eastern Alameda County and the cities of Pleasanton and Dublin, is also capable of generating a major earthquake (USGS, 2021). The last major earthquake caused by the Calaveras fault in Alameda County was an estimated magnitude 6.9 in 1861, which reportedly ruptured a six- to eight-mile-long stretch of the fault zone primarily through rural farmland and small towns. The entire population of Alameda County in 1860 was 8,927 compared to 1.6 million today.

The nearby San Andreas fault, which is shown in comparison to the Hayward fault in Figure 1.2, is one of the most active faults in the world moving at 5 to 7 centimeters per year (American Museum of Natural History, n.d.). The Loma Prieta earthquake was the last major activity along the San Andreas fault in 1989, recorded as a 6.9 magnitude. In 2008, the USGS estimated that over the next 30 years there is a 63% chance for a magnitude 6.7 or greater earthquake to occur in the Bay Area, likely to cost many lives and cause billions of dollars in damage (USGS, 2008). Earthquake shaking potential is predicted to be greatest along the Hayward fault causing surface fault rupture within Alameda County. The Bay Area is a hot spot for seismic activity, which can also trigger soil liquefaction and landslides. Figure 1.3 demonstrates the potential hazards of liquefaction and landslides in Alameda County. Structures that were constructed on top of or immediately adjacent to active faults, such as the Hetch Hetchy Aqueduct (which supplies drinking water to the City and County of San Francisco) crosses the Hayward fault in Fremont, as shown in Figure 1.3. Critical infrastructures such as this are typically designed to tolerate fault movement. However, numerous buildings, underground utilities and homes in Alameda County were constructed over the Hayward fault and may not have been designed to tolerate the impacts of fault rupture.

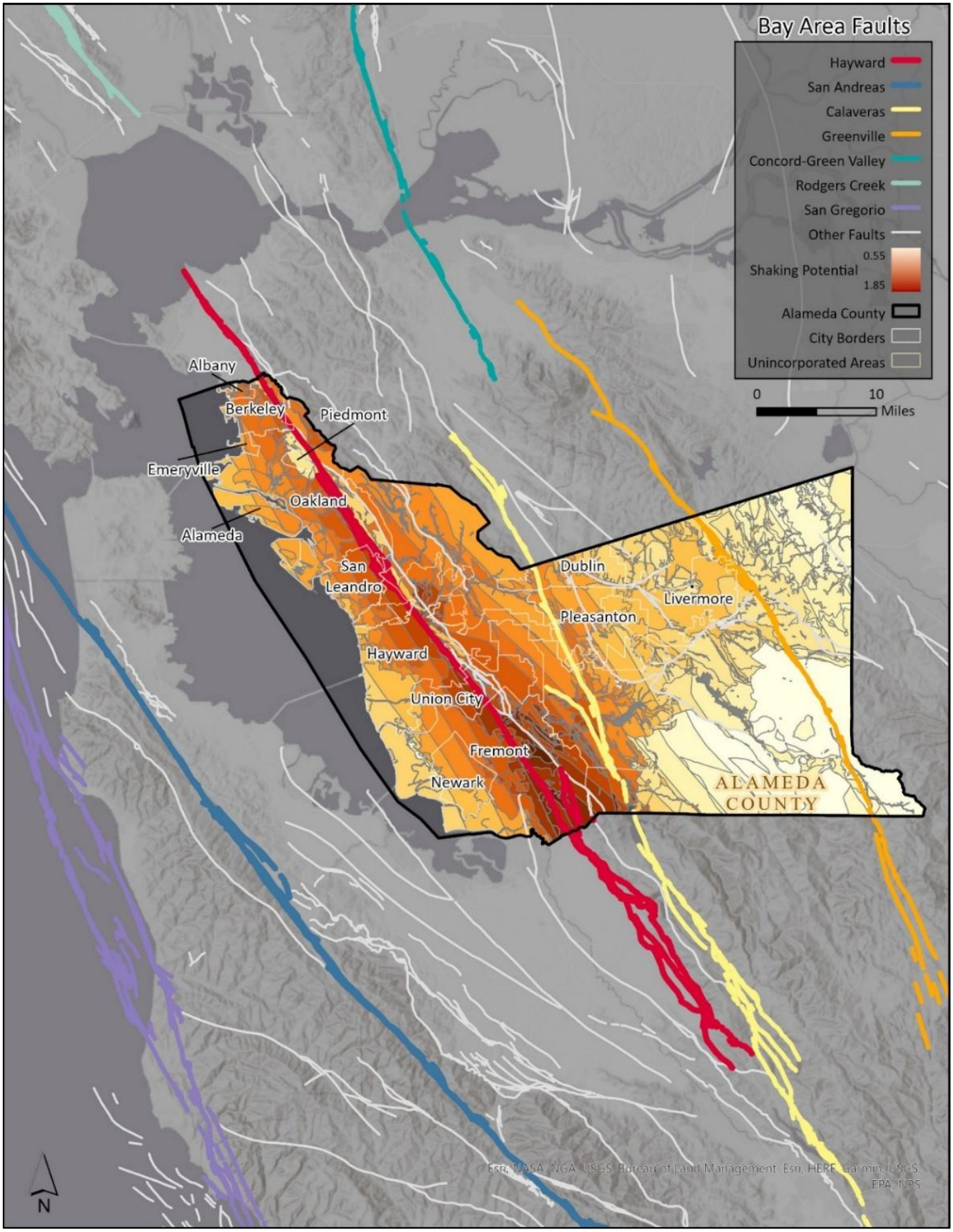


***Figure 1.2**: Bay Area Faults and Earthquake Shaking Potential in Alameda County*

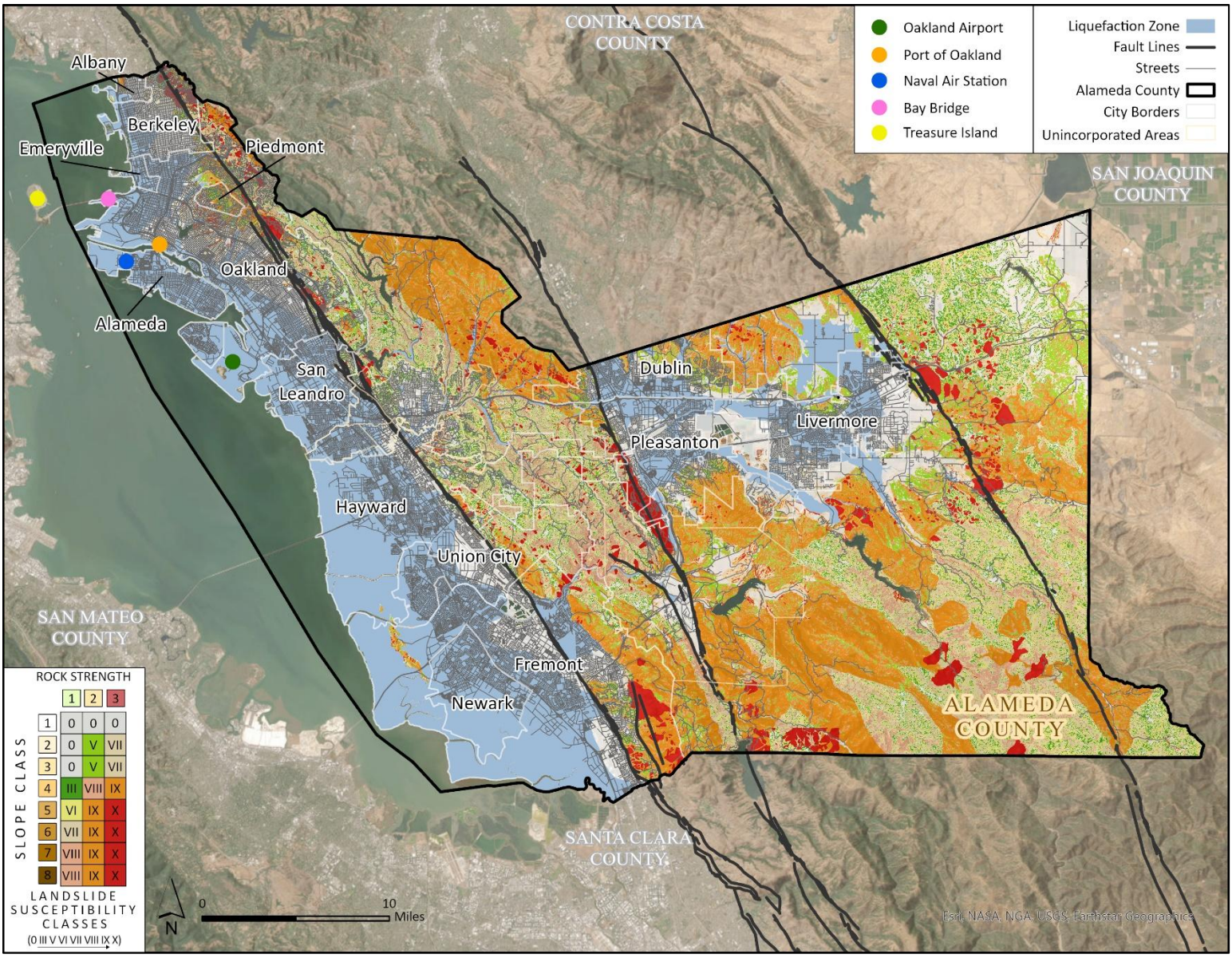


***Figure 1.3**: Landslide Susceptibility and Liquefaction in Alameda County*

Much of the land along the San Francisco Bay was built from dredge fill or was reclaimed land, meaning it used to be tidally influenced marsh. Both foundational methods are highly prone to liquefaction. Examples of developed areas built with bay fill are the Oakland airport, the Port of Oakland, the approach to the San Francisco Bay Bridge, Treasure Island, and the former Alameda Naval Air Station (Figure 1.3). Greater probability of liquefaction is caused by younger soils, closer proximity to the fault, and the depth of ground water. In addition to liquefaction, Alameda County must be prepared for landslide events. These occur along very steep areas with young bedrock and loose soil. Each hazard must be considered when planning mitigative action to reduce risk for people in the community. Figure 1.3 represents landslide susceptibility within the county. East of the Hayward fault is at high risk in regions labeled in orange and red. Dublin, Pleasanton, and Piedmont are each moderately to highly susceptible to earthquake-induced landslides and are located in liquefaction zones.

Population Sensitivity

Examining the 2020 Alameda County Local Hazard Mitigation Plan, nearly 100% of Alameda County's residents, single-point critical facilities, and flood control channels are in areas with severe to violent shaking potential, while areas with lower shaking potential are sparsely populated. Many of the critical structures in western Alameda County are located along the Hayward fault. In the event of an earthquake natural hazard, infrastructure like airports must be shut down for inspection to ensure the building's strength and runway safety before continued use. Vulnerable populations include residents of older homes, low-income communities, and elderly populations. Older homes have a high chance of serious damage or collapse in a higher magnitude earthquake. Buildings constructed before the 1970s would likely suffer more severe damage, putting any residents at greater risk. Communities with lower per capita income are more vulnerable because recovery after the hazard becomes exceedingly difficult. These residents might not have earthquake insurance, widening the gap between demographics during the recovery period. The elderly are more sensitive to hazard exposure because of possible health conditions or disability. If this population is reliant of specific medications or amenities, these might be difficult to obtain following a hazardous event. Sensitivity of a population is an imperative factor when determining high risk communities.

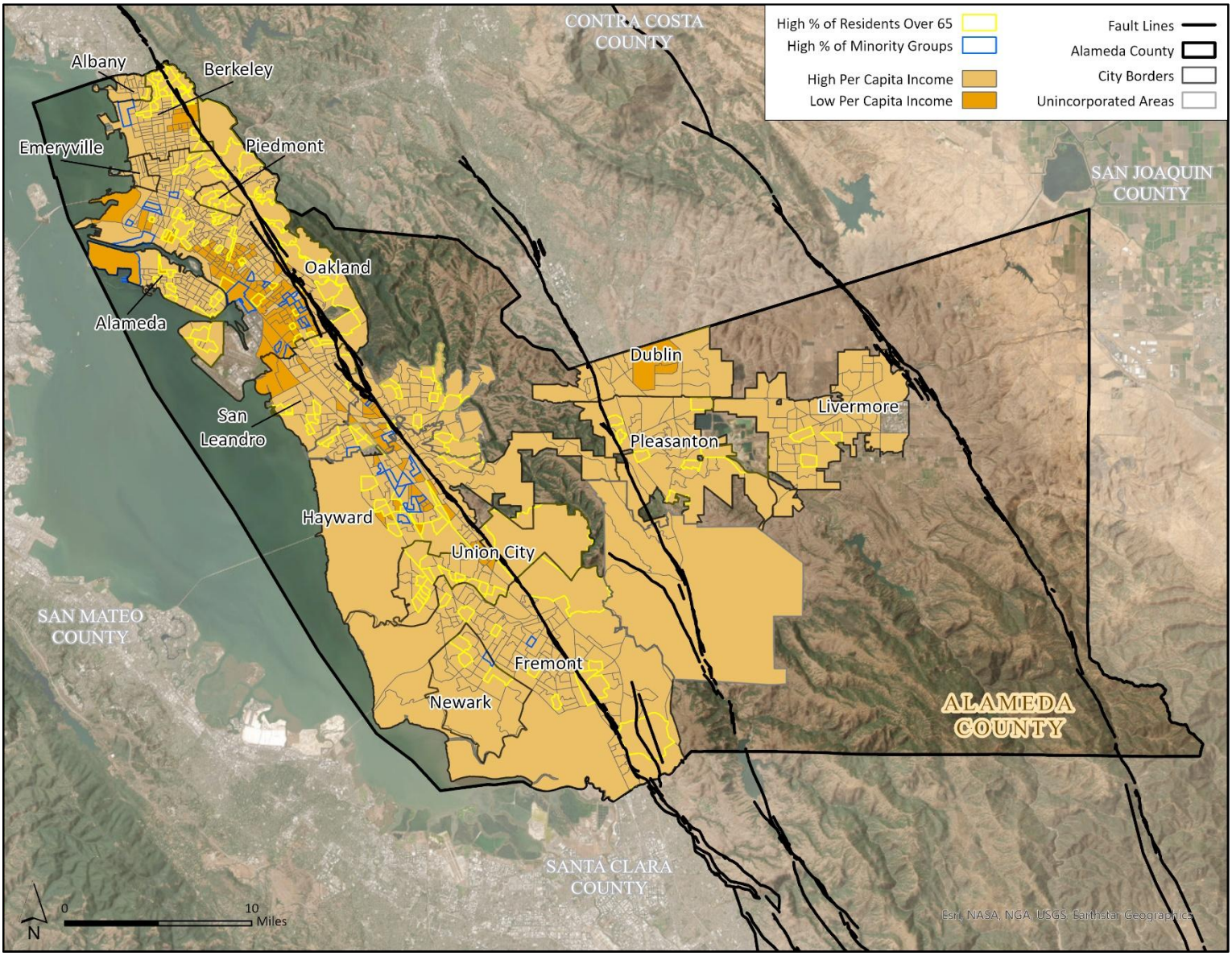


***Figure 1.4**: Vulnerable Communities of Alameda County*

Vulnerable Infrastructure

It is vital to understand how many people are located within high shaking potential zones, high landslide susceptibility areas, and liquefaction zones. According to the USGS, houses and infrastructure designed specially to withstand earthquake seismic activity could produce slight damage compared to ordinary or poorly constructed buildings which may sustain partial collapse or considerable damage. Populations living or working in these buildings are at a very high risk of serious injury or death in the event of a hazardous earthquake event. Proximity is one of many factors which calculate a person's risk to a natural hazard. Those who live in outdated housing and closer to the Hayward fault are at higher risk. The first earthquake codes to withstand horizontal forces were enacted in 1933 after a mild earthquake caused extensive damage in Southern California (Hamburger, 2022). Earthquake building codes were more thoroughly written in 1971 after a 6.1 magnitude earthquake in the San Fernando Valley (Hamburger, 2022). According to US Decennial Census data, an estimated 423,901 occupied housing units or 65.6% were built before 1979 in Alameda County (Table 1.2). The number of people within high-risk natural hazard areas will be analyzed in Chapter 3.

***Table 1.2**:* Occupied Housing Units by Year

Occupied Housing Units	Year Housing Unit Built						
	Before 1939	1940-1959	1960-1979	1980-1999	2000-2009	2010-2019	After 2020
# Occupied	120,714	112,897	165,471	113,143	41,103	43,268	11,938
% Occupied	19.8%	18.6%	27.2%	18.5%	6.8%	7.1%	2.0%

Economic Standing

Economic standing can play a pivotal role in the resilience of a community. The census block group data represented in Figure 1.4 shows the spatial relationship between low per capita income in Alameda County. The second metric analyzed for this Mitigation Plan is the percentage of low-income households in a high-risk area. There are 151 block groups across Oakland, San Leandro, Hayward, and Dublin experiencing low per capita income designated in dark orange. The distinction between low and high per capita income for these block groups was calculated by one standard deviation below the mean per capita income for the county. The Bay Area's average per capita income is $130,730 due to the wealth generated in the Silicon Valley compared to Alameda County's average of $47,692 (US Census Bureau Table, 2023). There is a large discrepancy between block group per capita income with the maximum of $160,391 and minimum of $3,695 (US Census Bureau Table, 2023). The residents of block groups below the mean income level will likely struggle to regain their footing following a hazardous event, especially if any valuable belongings were destroyed. Minority communities are often among the more vulnerable, Figure 1.4 outlines these communities in blue. A similar trend appears when comparing low-income and minority areas, the majority reside in South Oakland and Hayward.

Elderly Population

There is a high elderly population distributed across in Alameda County. Specifically, North Berkely in the hills, Piedmont, Oakland hills, Hayward, and Union City have a greater number of residents over the age of 65. It is vital to understand the distribution of these vulnerable communities in relation to the outlined high susceptibility hazard zones. According to the US Census Bureau 7% of the population in the County own or rent homes built before 1989 and are over the age of 65. Elderly people are at greater risk than a young person due to health factors or mobility. In the case of injury, a younger person is more likely to recover faster than a resident over 65 years old. The Berkeley and Oakland Hills are known for their large historical homes often occupied by older residents. This stereotype is displayed in Figure 1.4 as low-income regions rarely overlap with the elderly populations in these cities. Table 1.3 shows the percentage of the total 104,185 homeowners and 44,550 renters over 65 years living in homes built from before 1939 to after 2020 (US Census Bureau Table, 2023). This factor of sensitive populations will play a large role in the resilience of the community.

Table 1.3: Alameda County Elderly Residents by Building Year

Number of Residents	Year Structure Built									
	Before 1939	1940-1949	1950-1959	1960-1969	1970-1979	1980-1989	1990-1999	2000-2009	2010-2019	After 2020
Homeowner over 65 years	24.8%	8.4%	13.9%	12.4%	13.7%	12.6%	7.2%	4.7%	1.8%	0.2%
Renter over 65 years	21.6%	5.8%	9.6%	12.6%	16.9%	9.2%	6.5%	7.8%	8.6%	1.1%

Adaptive Capacity

Alameda County has prepared three Local Hazard Mitigation Plans starting in 2010, 2016 and the most recent in 2021. Past earthquakes in the area, like the 1898 earthquake along the Hayward fault, killed five people but destroyed every building in the small town of Hayward at the time (Perkins, 2010). In comparison, the Loma Prieta Earthquake of 1989 killed 63 people and caused $6 billion in damages (Perkins, 2010). The significant increase in population and infrastructure of the 21st century will inhibit and complicate the mitigation process. There is a saying that “earthquakes don't kill people, buildings do,” signifying the importance of properly constructed buildings. Building and seismic codes are a method to reduce this risk. The International Code Council (ICC) maintains the model building codes, although some local jurisdictions adopt specified codes for the region. FEMA’s Earthquake-Resistant Design Concepts explicitly states building codes do not require existing buildings to adhere to current code requirements.

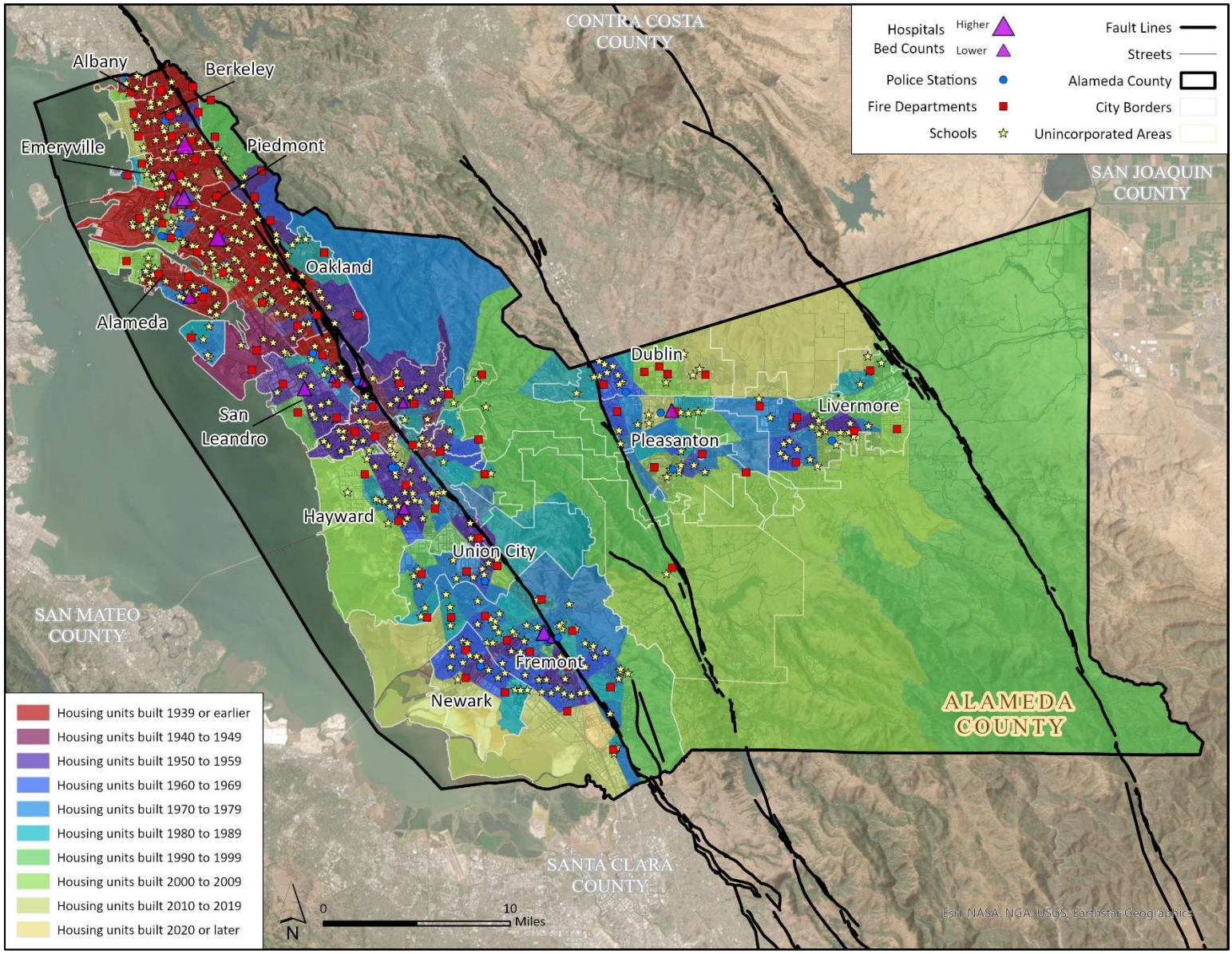


***Figure 1.5**: Critical Facilities of Alameda County*

Each mitigation plan created for Alameda County addressed mitigative strategies. The 2016 plan implemented a new education-based response plan. The county was tasked with investigating potentially vulnerable public and private utility systems including sanitation/sewer, and fuel pumping stations (Local Hazard Mitigation Plan, 2016). They developed and implemented plans to increase building owner's general knowledge and appreciation of seismic upgrading for a building's structural and nonstructural elements. (Local Hazard Mitigation Plan, 2016). The HayWired Earthquake Scenario calculated with Alameda County's 413,505 buildings, are worth 155,699,818 (thousands of dollars) in total exposure value (Detweiler et al., 2022). When constructing a new building an additional 1% in costs is added by reaching a "functional recovery standard". Following a major earthquake, 75-95% more homes would be structurally sound and available to use (Detweiler et al., 2022). The 2020 plan acknowledges the importance to seismically retrofit or replace County and local ramps and bridges (Local Hazard Mitigation Plan, 2020). The mitigation plan for earthquakes was to retrofit or upgrade existing vulnerable critical facilities to protect structural and nonstructural components (Local Hazard Mitigation Plan, 2020) but fails to incorporate the importance of reinforcing residential buildings.

Fremont, Hayward, and Union City face the highest predicted shaking and fault rupture from a Hayward fault earthquake. Figure 1.5 displays infrastructure across the county to demonstrate spatial proximity to the fault. Highway 580 runs almost directly parallel to the fault line causing severe damage in the case of fault rupture. Critical aging pipes in regions of extreme shaking hazard and high landslide susceptibility were planned to be replaced. Hospitals, police officers, and firefighters will be in high demand following a major earthquake, so their distribution across cities will be beneficial. In Figure 1.5 hospitals with more beds are represented in larger purple triangles. Children are more vulnerable in a natural hazard event so it is important to understand the spatial relationship of schools to the greater earthquake shaking potential represented in Figure 1.2. These critical facilities are generally evenly distributed, but many are located along the fault lines.

Conclusion

Alameda County is at high risk of experiencing an earthquake based natural hazard in the next 30 years. Due to the dense population of the region and proximity to major fault lines, this community has high exposure to this hazard. Specific communities within Alameda County are more vulnerable to earthquake based natural hazard due to the wide range of per capita income, age, and ethnicity demographics. The adaptive capacity of Alameda County was thoroughly thought out for mitigation and response plans but does not consider equitable resilience. Equity is the understanding of the moral equality of people (Matin et al., 2018). Overlooking disadvantaged communities during formation of mitigative strategies sorely misrepresents the population's needs. MacKinnon and Derickson (2013) and Fainstein (2015) contend that indifference to resilience may favor existing privileged social relations (Matin et al., 2018). Failure to incorporate vulnerable communities into the existing 2020 Local Hazard Mitigation Plan for Alameda County, is a significant oversight in the planning process. There was thorough consideration for city infrastructure like bridges and pipes, which are evidently important. However, with the existing plan the County cannot promote safety for all residents.

Communities in Oakland and Hayward appear to possess more of a risk due to their combined exposure, sensitivity, and adaptive capacity. There is a larger population of low-

income households, minority communities, and people over the age of 65. The combined impact of fault rupture, landslides, and liquefaction will significantly displace the community. Whether or not a household has the financial means or resources to quickly remedy the situation determines a community's vulnerability. Existing mitigation plans for Alameda County do not incorporate the risk to the people rather than the infrastructure. This is an important aspect but the people at risk are the most vital. Utilization of census block groups to analyze the entirety of Alameda County may develop limitations in understanding the full nature of each city's most vulnerable communities. Census blocks would form a more complete picture of the needs for individual people. Grouping all minority ethnic groups together for analysis purposes limits the awareness of specifically vulnerable ethnicities. Educated assumptions were drawn about the parameters for most vulnerable communities in Alameda County, metrics like disability status, households with children, or education level may have developed better estimations for accumulated risk. To provide equitable resilience to these vulnerable communities, stakeholders and policy makers need to incorporate the policies outlined in Chapter 2.

Goals and Policy Objectives

The hazard, population sensitivity, and adaptive capacity in Alameda County outlined in Chapter 1 form the basis for the mitigation plan. To mitigate the harmful effects of this hazard, the main goal of Alameda County's policymakers should be to reduce damage to infrastructure of vulnerable communities. An earthquake greater than magnitude 5.0 will cause catastrophic damage across the Bay Area. The dense populations and quantity of old infrastructure will make recovery from an event more challenging. The impacted communities may receive fewer resources due to their demographics, income level, or even age. Chapter 2 introduces two policies to combat these effects. Resources provided during the policy programs should include education on hazard preparedness and recovery. Aid after a disaster may be prioritized for more outspoken communities. A greater understanding of risk, along with more insurance experience creates a greater desire for housing earthquake insurance (Zhang & Qian, 2018). Proper insurance one facet needed for a quicker recovery after a natural hazard.

GOAL 1 – Reduce hazard impact: Existing Structure Retrofit Program (ESRP) Control the impact on human lives through modification of existing foundations and construction methods.
POLICY OBJECTIVE – Reinforce existing occupied buildings. Upgrade the foundation of aging buildings.
RATIONALE – 65.6% of the total population live in housing units built before 1980. Older housing is likely not up to the current earthquake building code. For the safety of the community, it is important to identify these buildings and the people who own or rent the space to provide education on the issue. Lowering damage risk will reduce costs after an earthquake to repair this damage. Analysis in Chapter 3 will determine the most vulnerable houses by age, predicted proximity to extreme shaking during an earthquake, liquefaction zones, and high landslide susceptibility areas. Following the selection of the most vulnerable structures, residents will be informed of the safety hazard and be provided with the opportunity to modify the existing structure. If reinforcement is requested, relocation to predetermined group housing will be provided, if required, until construction is complete. These steps will provide equitable resilience by reaching all facets of Alameda County's community. Retrofitting an existing property in Alameda County can range from $14,000-$27,000. To reduce the divide between income classes following a natural hazard event this service will be offered to anyone regardless of income. Funding will be obtained through local and federal grants and subsidies. The ESRP will be provided in tandem with the recently developed Earthquake Brace + Bolt (EBB) and Earthquake Soft-Story (ESS) retrofit programs. EBB and ESS offer California homeowners money to pay for upgrades to the foundation of the home. By reaching out to the most vulnerable houses and providing thorough education, more of the Alameda County community will be reached. This action will save countless lives since old or unreinforced buildings are at high risk of collapse in the event of a major earthquake.

GOAL 2 – Reduce hazard Exposure: New Affordable Housing Program (NAHP) Reduce exposure to earthquake hazard by changing a community's proximity to the hazard or by taking mitigative action to reduce the severity felt during the event.
POLICY OBJECTIVE – Build new affordable housing focused on earthquake code parameters. Relocate vulnerable communities into reinforced housing.
RATIONALE – New multi-family housing units will be constructed in an area with medium-to-high suitability. These housing units will be constructed with the most sensitive populations of Alameda County in mind. Advertisements for the housing will be primarily distributed in low-income communities and available in any language. The vulnerable communities are defined in Figure 1.4 as residents who are low-income, elderly or live in older housing. Residents of older buildings will be primarily targeted for relocation. To encourage relocation, compensation will be provided. Local construction companies and contractors will be hired to ensure thorough knowledge of the proposed site. Proper construction of the foundation of the buildings will be a focus, as many of the current residential areas are built in potential liquefaction zones. Out of the two proposed policies, NAHP would be more costly due to the full construction of new housing. Funding will be obtained through grants and governmental subsidies. Affordable housing is a priority as most housing projects in the Bay Area do not focus on helping vulnerable communities. Existing buildings will be repurposed for non-residential use. As many vulnerable communities currently live in high-risk areas it is important to focus on improving their adaptive capacity.

Methods

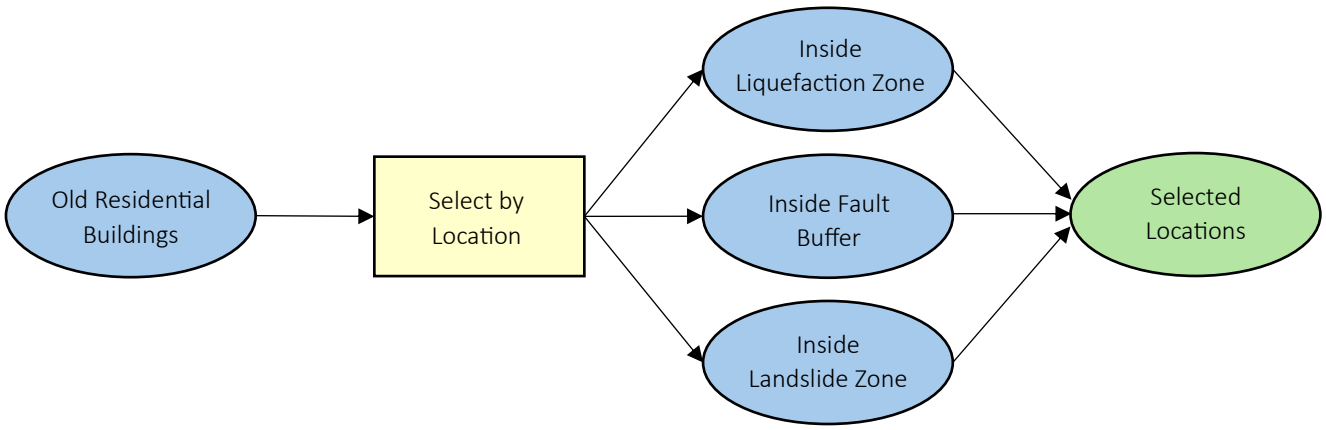


***Diagram 2.1**: Identify Most Vulnerable Housing for Retrofitting*

Diagram 2.1 addresses Goal 1 outlined in Chapter 1 to reduce the hazard impact. The diagram shows the path of analysis for the old residential building layer to select which buildings require retrofitting. The Select by Location tool was incorporated to find buildings inside of the potential liquefaction or landslide zones or within 1000 meters of the Hayward fault. The Buffer tool was used to create circular polygons 1000 meters around the Hayward fault. The Year Built layer was used for the selection to find the most vulnerable locations to destruction in the event of an earthquake. These selections were represented by white stars seen in Chapter 3 Figure 3.7-8. The output of this selection provided only residential buildings, which would benefit the most from retrofitting. The Year Built layer focuses on housing older than 1950 due to the higher potential danger in the event of a natural hazard. The first mandatory seismic codes for the State of California were published in 1933 after extensive damage was caused by the Long Beach Earthquake. The year-built data was collected from the publicly available data from the real estate company Redfin and historical society data. Redfin caused a limitation in finding the oldest existing houses due to filters excluding houses older than 1900. The data found from the Oakland and Berkeley Historical Society may not be fully representative of each city's historical housing. Despite these limitations, highly vulnerable communities are more likely to receive aid from this program. Therefore, the selected housing will be analyzed in Chapter 3 to determine overlap with the vulnerable communities.

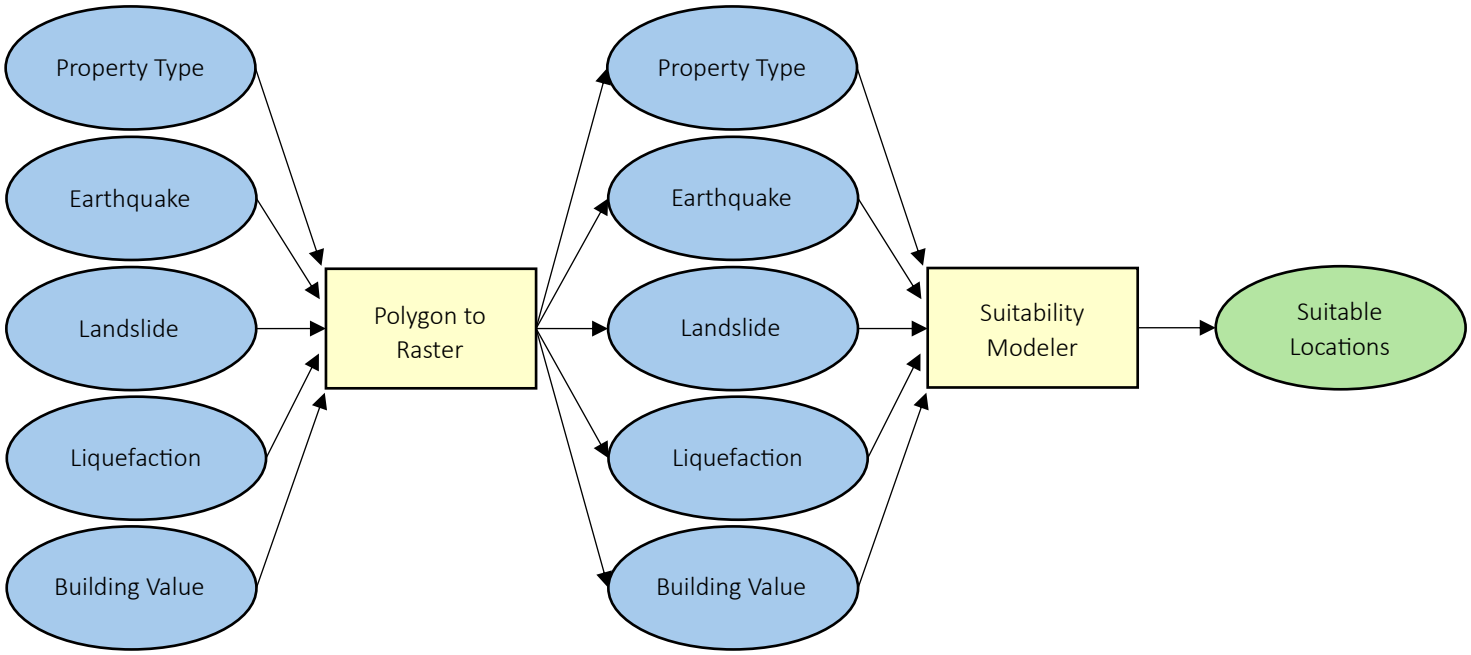


Diagram 2.2: *Suitability Analysis*

Diagram 2.2 addresses Goal 2 to reduce hazard exposure. The diagram visualizes the Suitability Modeler analysis to understand what regions will experience the highest danger in the event of an earthquake natural hazard and where are the most suitable locations for housing relocation. To relocate vulnerable communities, areas of high suitability need to be established. The earthquake shaking potential layer from Figure 1.2, landslide susceptibility and liquefaction zone layers from Figure 1.3, and finally new building value and property type layers were all used for this analysis. These natural hazard layers were used for analysis because these are the three main hazards associated with earthquakes. Each vector layer was geoprocessed using the tool Polygon to Raster ensuring compatibility with the Suitability Modeler. A scale of 1 (least suitable) to 3 (most suitable) was implemented for each layer. For landslide susceptibility, a score of 3 was given to level 0, a score of 2 for level 3, and a score of 1 for the remaining categories. The same process of suitability selected was applied to the remaining layers, making the most dangerous areas 1 and safer areas 3. Since the suitability scores were applied manually, assumptions were made on the scale of suitability for each parameter. After each layer was run through the modeler, the locate feature was utilized to find the most suitable location for new affordable housing units to be constructed. The Suitability Modeler was used separately in the cities of Berkeley and Oakland. Only ten sites were chosen in each city for construction because there are few suitable locations for entirely new buildings in these cities. The sites for relocation were selected using the Select by Attribute and Location tools to highlight the oldest residential buildings, old buildings in low-income areas, and elderly communities. Lines were drawn connecting the selected building sites to the 1-5 closest selected relocation sites. To perform this analysis on additional Alameda County cities, the same methods outlined above are required.

The results of the geospatial analysis described in Chapter 2 are provided below. Analysis of the Existing Structure Retrofit Program (ESRP) and the New Affordable Housing Program (NAHP) against the baseline conditions set in Chapter 1 and cost-benefit metrics are depicted in Table 3.4. The focus for the analysis was the City of Berkeley and Oakland because both cities appeared to have a greater quantity of old residential buildings and vulnerable communities than other cities in the county. The analysis methods outlined in Chapter 2 can be applied to the remaining cities of Alameda County to provide equal aid to sensitive populations. A map for each selected city implementing each policy goal and respective analysis is discussed in this chapter.

***Table 3.4**: Policy Objectives Analysis*

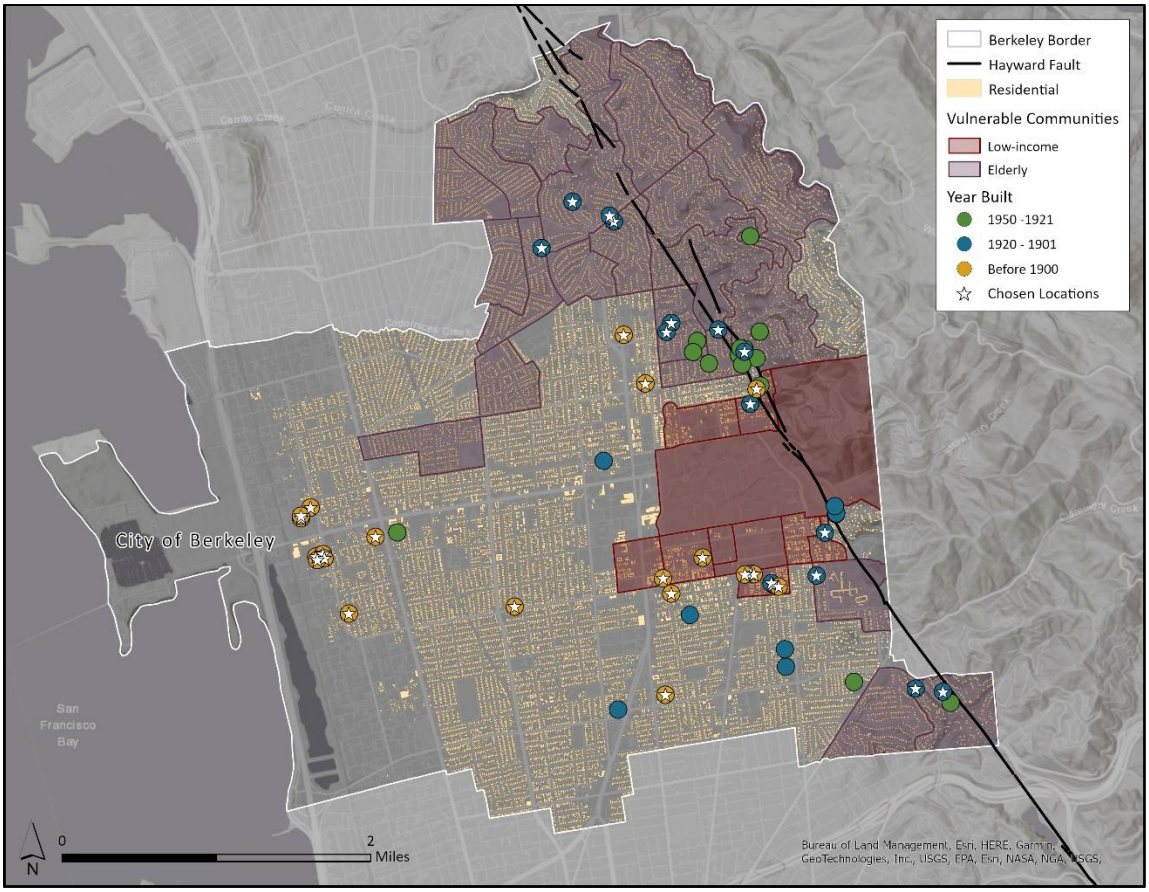
	Metric 1	Metric 2	Metric 3	Cost Metric	Benefit Metric
Baseline Conditions	Residential Infrastructure: 62 houses in Berkeley and 65 houses in Oakland built before 1950	Economics: 10 low-income block groups in Berkeley and 106 in Oakland, below the county average of $47,692	Age: 23 elderly block groups in Berkeley and 41 in Oakland, over the age of 65	Reduced safety and long recovery time following natural disaster	For Alameda County these three metrics are most applicable
Policy #1 – Existing Structure Retrofit Program (ESRP) Reduce Hazard Impact	Reinforce existing housing: 106 total old houses in Berkeley and Oakland; 37 selected in Berkeley, 19 selected in Oakland	11 houses in Berkeley and 2 houses in Oakland in low-income block groups	11 houses in Berkeley and 3 houses in Oakland in block groups populated by elderly residents	Cost to reinforce 56 buildings in vulnerable communities: average $12,500 per building; total of $700,000	Avoided damages to properties, vulnerable communities emphasized
Policy #2 – New Affordable Housing Program (NAHP) Reduce Hazard Exposure	Construct 10 new up to code housing units in Berkeley and Oakland and advertise to current residents of houses built before 1900	Strive for affordable housing: 11 Berkeley and 4 Oakland houses in low-income block groups targeted for relocation	Advertisements for sensitive populations: 11 Berkeley and 5 Oakland houses in elderly population block groups targeted for relocation	Cost to build 10 new affordable multi-family housing units in each city: average $665,000 per unit; total is $6,665,000 per city	Fewer casualties, aid vulnerable communities in reducing exposure to hazard

Policy Objective 1

Reducing the negative impacts on vulnerable communities caused by a natural hazard is the focus of the first proposed policy. The Existing Structure Retrofit Program (ESRP) will retrofit the existing foundations of buildings constructed before the 1950s to improve the resilience of communities. Disastrous events do not discriminate in causing destruction, but individual families or communities may suffer disproportionately after natural hazard events (Cooley et al., 2012). This inequality is evident in disaster preparedness, response, and recovery. Social vulnerability is visible in low-income and elderly communities as recovery is often more difficult than the average household. Low-income populations are less likely to be insured or likely under-insured (Cooley et al., 2012). In addition, they are more likely to live in a lower-quality constructed home which may be more susceptible to damage (Cooley et al., 2012). After a hazardous event, elderly populations may need more medical assistance, creating higher vulnerability. A limitation of ESRP is the underrepresentation of homes built before 1950, as discussed after Diagram 2.1 due to limitations caused by the Redfin search system. The distribution of older buildings was analyzed in Figure 3.7-8. Berkeley and Oakland demonstrated a cluster of old residential buildings near the waterfront. Low-income and minority communities in Oakland live closer to San Francisco Bay due to redlining. This is the region most prone to liquefaction and in the maps created by the Homeowners Loan Corporation (HOLC) in 1937 the land near the bay was deemed “lower” and “hazardous” (Susaneck, 2024). The properties there were predominately sold to Black, Asian, and Hispanic people while the “best” or “desirable” land was withheld for white people (Susaneck, 2024). These decisions built generational wealth for the white homeowners, leading to the division in vulnerability today.

Figure 3.6: *Redlining in Oakland (Cooley et al., 2012)*

After identifying old residential buildings, 37 were selected in Berkeley and 19 in Oakland for ESRP. Buildings were selected based on age and distribution within vulnerable communities. If a house was built before 1900 no matter the location, it was chosen. If a home was built before 1920 and located in a low-income or elderly block group, then it was also selected. Of these chosen buildings in Berkeley, 11 were in the low-income block groups and 11 in the elderly block groups. In Oakland, two buildings were in the low-income block groups and three in the elderly block groups. The California Residential Mitigation Program (CRMP) states the average cost of an earthquake retrofit is between $3,000 to $7,000 for the Brace + Bolt retrofit (2025). A soft story retrofit meaning a living space exists over an attached garage can cost between $15,000 to $25,000 (CRMP, 2025). The average cost of these estimates is $12,500 which will be used to calculate the total cost of $700,000 for implementing ESRP in Berkeley and Oakland. Funding will be obtained through grants and governmental subsidies. Focusing restorations on vulnerable communities reduces the gap in preparedness and recovery between less sensitive groups, by reducing damage to properties.



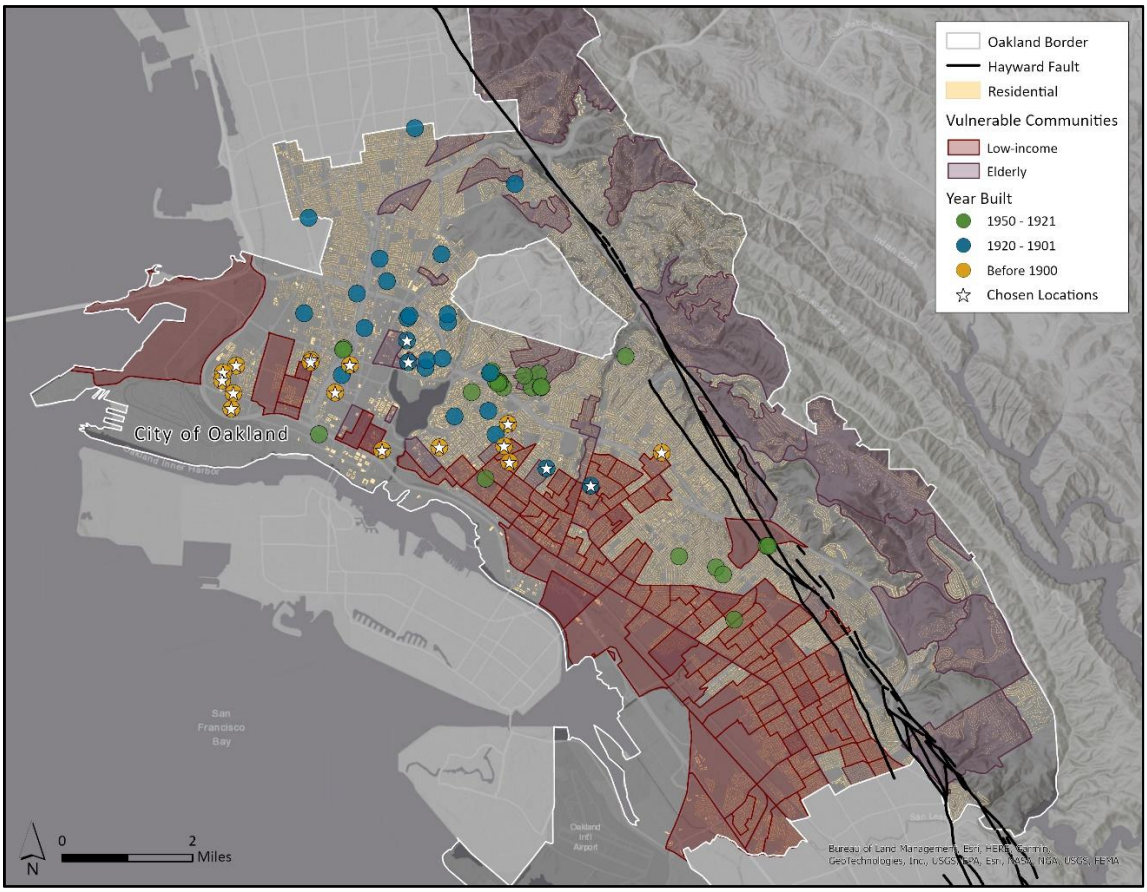
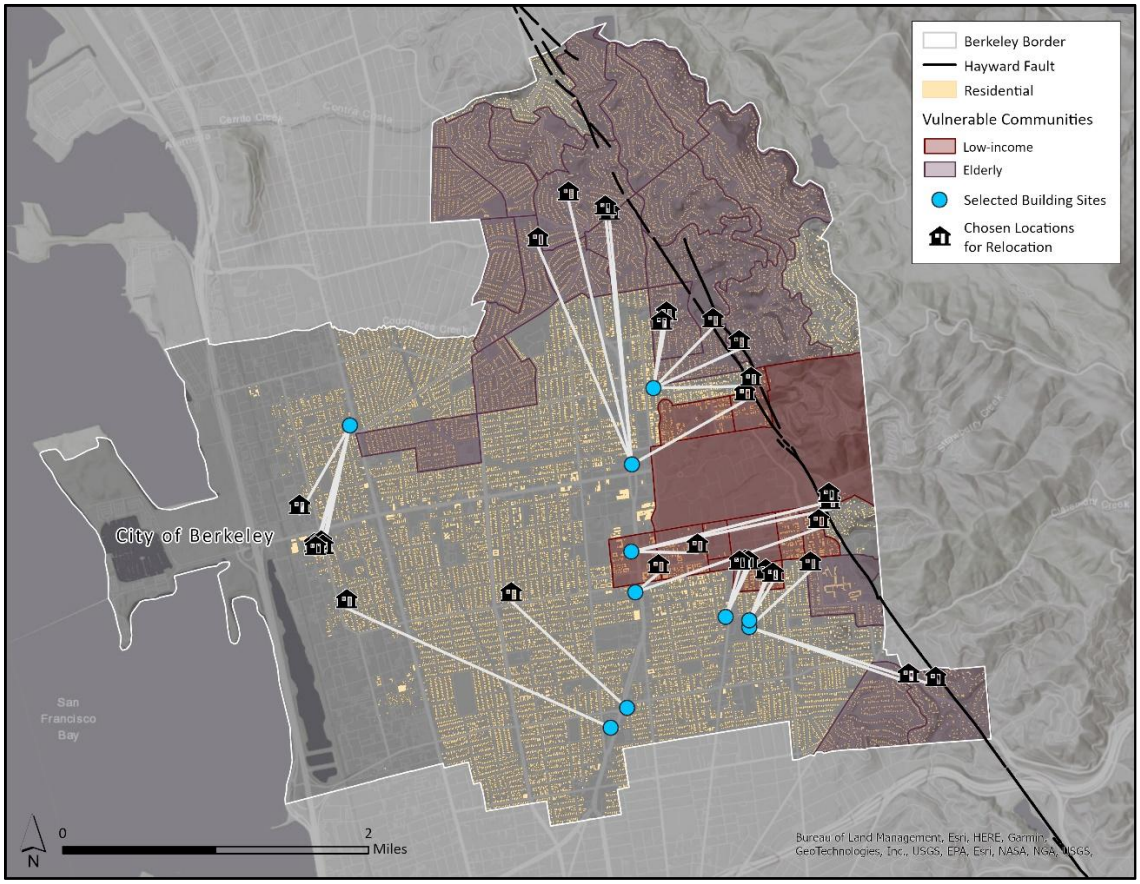


***Figure 3.7-8**: Policy #1 – Reduce Hazard Impact Analysis in Berkeley and Oakland*

Policy Objective 2

The New Affordable Housing Program (NAHP) will reduce earthquake exposure for vulnerable communities. Relocating vulnerable households to new multi-family housing units will lower risk. The Suitability Modeler was used to select the best locations for a new construction project. Vacant and unclassified building types were used for the most suitable locations so existing buildings would not need to be torn down. Ten new sites were chosen in each city for the proposed housing units, as seen in Figure 3.9-10. Due to the limited suitability of vacant lots, only ten sites were chosen to ensure proper construction using the strictest earthquake building codes. Costs in the Bay Area for multi-family housing are $400 to $625 per square foot compared to single-family homes are $330 to $550 (Landes, 2024). Despite the higher cost, more sensitive populations can be accommodated. The focus of this proposal was to emphasize the construction of affordable housing without limiting building safety. Again, residents of old buildings, low-income communities, and elderly populations were targeted for NAHP. Similar parameters in ESRP were used to select which housing units would be targeted for relocation. All residential buildings built before 1900, located within low-income or elderly block groups were selected for the relocation program. A similar pattern observed in both cities shows that elderly populations are concentrated in the hills. This trend aligns with the general understanding that the Berkeley and Oakland hills have large homes primarily occupied by older white residents. This demographic pattern reflects the historical impacts of redlining, as outlined in Figure 3.6.

A comparative program called Homekey began in 2020 which has built more than 400 permanent supportive housing units for formerly homeless people (Angust & Fagan, 2024). Additional funding was received in 2022 through a $200 million bond to build more affordable housing in Oakland (Angust & Fagan, 2024). Building upon the Homekey program, NAHP would focus on advertising for the housing locations shown in Figure 3.9-10 to promote equitable resilience. The average cost of constructing multi-family housing units is $665,000 per unit, totaling $6,650,000 for ten units per city (Economic Institute, 2019). NAHP would be funded through public subsidies, with the properties designated for low-income or vulnerable residents. To promote relocation, part of the grant money will be allocated as compensation to the residents who choose to relocate. The overall costs come from the purchase of land and acquisition, construction, contractor fees, permanent financing, legal fees, reserves, developer fees, and commercial costs (Economic Institute, 2019). Construction costs in the Bay Area are the highest in the United States but have been slowing in their increase since 2021 (Landes, 2024). Fortunately, following the pandemic, subcontractors have become more available due to the drop in active projects, which has sped up construction time and led to higher-quality work (Landes, 2024). If construction became feasible, relocating vulnerable communities would help minimize casualties during natural hazard events by reducing their exposure.



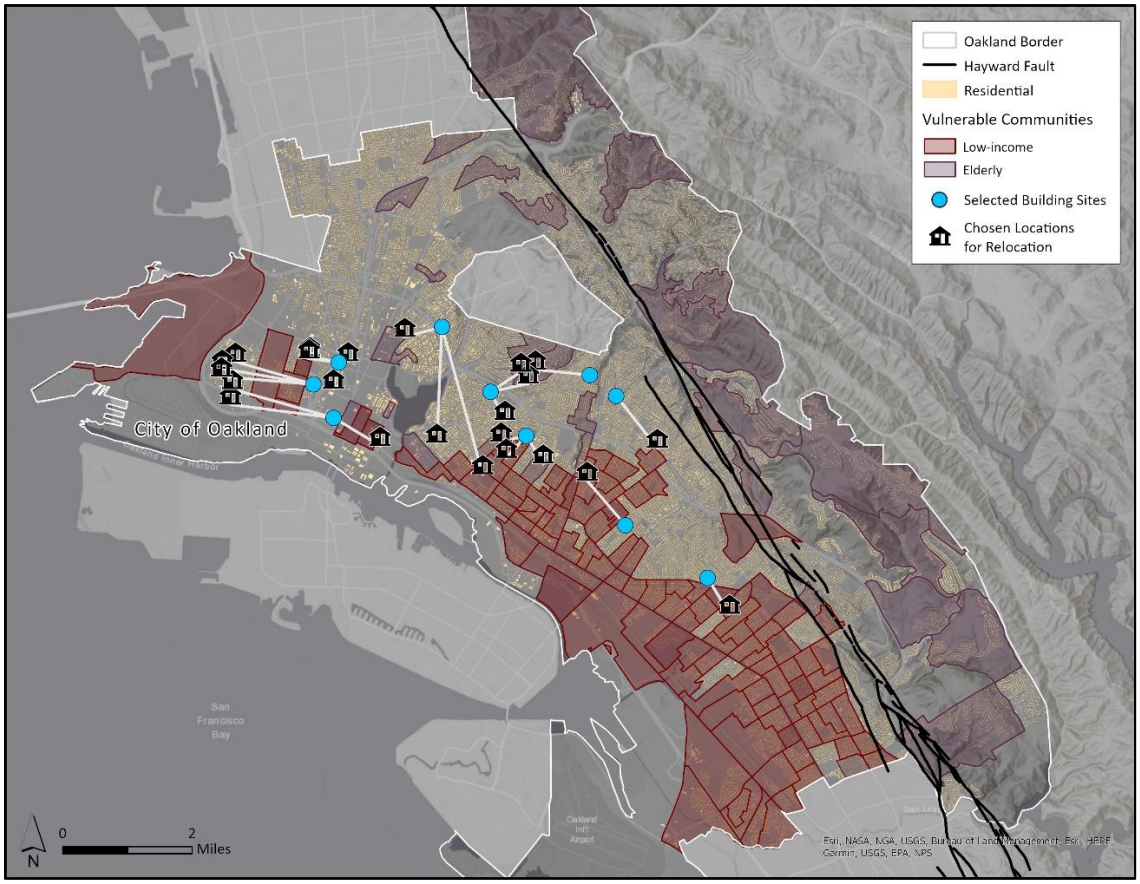


***Figure 3.9-10**: Policy #2 – Reduce Hazard Exposure Analysis in Berkeley and Oakland*

Policy Objective Recommendations

Based on the analysis in Chapter 3, the Existing Structure Retrofit Program (ESRP), is recommended to implement into Alameda County's Hazard Mitigation Plan. The damage caused by an earthquake would be greatly reduced for old residential buildings occupied by vulnerable communities. Vulnerable communities like low-income and elderly populations may have access to fewer resources before or after natural disasters causing vulnerability. Residents of old buildings are at a higher risk of property destruction during a large-scale earthquake. By restoring existing buildings, costs will be reduced compared to the second proposed project which constructed new affordable housing up to earthquake code. Retrofitting is a faster process than constructing a new building so a major cost of the New Affordable Housing Program (NAHP) is the extended timeline. According to the USGS, a major quake is likely to occur at fault in the Bay Area by 2032 (USGS, 2008). This urgency to prepare vulnerable households is a key justification for the retrofitting program. The program will build upon the existing California Residential Mitigation Program (CRMP) to focus time, effort, and costs on households who are low-income, elderly, and live in a home built before the 1950's. The focus of ESRP is to provide equitable resilience to the underserved communities of Alameda County (Matin et al., 2018).

Oakland, Berkeley, and Fremont have the highest population of homeless people out of the cities of Alameda County. As of 2024, Oakland had 5,490 people, Berkeley 844 people, and Fremont 809 homeless people according to the San Francisco Chronicle (Angust & Fagan, 2024). Across the county, 9,450 people are homeless. Although these people could not currently benefit from the retrofit program, these numbers depict the crisis of rising housing costs and the stark divide between the maximum and minimum per capita income of $160,391 to $3,695 (US Census Bureau Table, 2023). Low-income residents likely could not afford to make these updates to their homes, raising the likelihood of damage in an earthquake. Costs following a natural hazard event would be detrimental and likely the number of homeless people in the county would rise. Implementing this program in vulnerable communities will increase their adaptive capacity (Engle, 2011).

The NAHP would require relocating people, a challenging process both logistically and in persuading individuals to move, even to a safer home. So, by utilizing ESRP instead, families can stay in their familiar neighborhoods but be comforted that their house is better adapted to a hazardous earthquake, during liquefaction, or from a landslide. Since a similar program already exists, it will be easier to gain funding for the project through grants and government subsidies. This adaptation will avoid maladaptive practices by improving infrastructure for vulnerable communities (EarthScan, 2024). Shifting vulnerabilities could occur for a community like minority groups who are unrepresented in ESRP-targeted vulnerable groups. This could cause decreased adaptive capacity for these excluded communities. An additional maladaptation is the longevity of the program. Even with a strengthened foundation and following the

earthquake building code, these updates may become less effective as time progresses, again increasing risk to the residents. To combat this, revisiting properties addressed in ESRP and updating the property further after 10-20 years is necessary to alleviate this risk. A benefit of ESRP is the elimination of complications related to land use planning. Conflicts may have arisen in NAHP during the selection of property to build despite considering property type during the Suitability Modeler discussed in Chapter 3.

Next Steps and Indicators

To measure the success of the Existing Structure Retrofit Program (ESRP) to reduce the impact of natural hazards, community decision-makers can record the extent of buildings retrofitted, compared to the known sensitive populations outlined in Chapter 1. Since analysis was conducted only for the cities of Berkeley and Oakland, replication of methods outlined in Chapter 2 will be necessary. The following steps outline indicators for success and methods for program completion.

Steps to move forward

1. Repeat analysis for remaining cities of Alameda County.
2. Identify most vulnerable households during this analysis.
3. Reach out to households and provide education on earthquake shaking potential, liquefaction zones, and landslide susceptibility.
4. Survey households selected for retrofitting program to understand willingness to participate and baseline sensitivity metrics.
5. Using grants and subsidies acquired from federal funding, hire local companies to perform retrofit.
6. Survey households following update of infrastructure to understand perception of safety and anxiety of natural hazards.
7. In the following 10-20 years revisit participating households to conduct analysis of strength of house and provide additional updates as needed.

Indicators

1. Less anxiety surrounding natural hazard events.
2. Public meetings are organized in each city in the county to educate the public on natural hazard preparedness, recovery, and ESRP. Advertise in multiple languages and provide incentives to attend for largest participation.
3. High community engagement signals preparedness and potentially decreased risk to the population.
4. Decreased destruction of property following a natural hazard event.
5. Recovery time of sensitive communities is about equivalent to that of non-sensitive communities.

Conclusion

Alameda County is at high risk of experiencing a major earthquake in the next ten years, so there is an urgency to prepare the county for this natural hazard. Not only will the ground shaking caused by the initial earthquake produce extensive damage to communities, but liquefaction and landslides often result from the shaking, resulting in additional damage. Focus should be given to sensitive populations like low-income, elderly, and residents of old infrastructure. Low-income residents may experience unequal financial recovery times following an event, elderly populations are at higher risk of health complications when many hospitals are overwhelmed with injuries caused by natural hazards, and any resident of an old building is more likely to experience the destruction of property during an earthquake. Populations like minority groups are often more sensitive to natural hazards so additional analysis may need to be conducted to select more vulnerable housing for the program. Due to these vulnerabilities, a policy was proposed to reduce these impacts. Retrofitting the houses of these vulnerable communities would greatly reduce the impact following an earthquake. An additional policy to identify suitable locations for the construction of affordable housing was proposed to relocate sensitive communities, however, ESRP is recommended to community decision-makers. There is a high distribution of low-income and old buildings along the Bay Front which is the zone most susceptible to liquefaction. This region was subjected to redlining in the 1930s and this decision is felt by the existing communities today. The unequal distribution of sensitive communities makes ESRP a high priority for this region. Alameda County needs to instate equitable resilience to ensure equal recovery after a natural hazard event.

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